Real Polytypes and the Hexagonality Law

Silicon carbide, silicon, germanium and cerium against the straight line

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. The Hexagonality Law (#66) says that in ideal stackings every kind of piece totals a straight line in the share h of hexagonal layers. DICTO proposed to test it against real crystals. In the language physicists use for stackings, the axial Ising (ANNNI) model, we prove that a property is a straight line in h exactly when its couplings between layers two and three apart vanish, J2 = J3 = 0, and that its stacking faults then stand intrinsic : extrinsic : twin = 2 : 2 : 1, the ratios of #67. We then read published data. The computed volumes of the silicon carbide polytypes lie on the line within the precision of the published numbers; the measured ones scatter between laboratories by more than the whole effect. Silicon and germanium bend slightly but clearly, by about 0.04% of the volume: their volume couplings J2 are of the size of J1, a relaxation that ideal geometry does not have. Germanium's stacking-fault energies come close to the ideal 2 : 2 : 1 (2.25 : 2.06 : 1), silicon's do not (4.1 : 2.4 : 1). Silicon carbide's band gap is far from straight. In cerium, the double-hexagonal form at h = 1/2 has 1.2% more volume than the cubic form. The straight line is the ideal part of every stacking; the departures measure what real crystals add.

1. From the ideal to the real

The Hexagonality Law was proved for the pieces of ideal stackings, where space is shared by weight among atoms and holes at fixed positions (#66), and #67 drew its consequences for random stackings and faults. DICTO proposed the next step: to look at real crystals. Real crystals relax: their layers move apart or together a little, and their atoms shift. What matters is how close real stackings come to the straight line, and what the departures mean. We use only published numbers, and say for each how precise it is.

2. The straight line in the physicists' language

A stacking is a sequence of steps $\sigma_i = \pm 1$ from layer to layer (Hägg); a layer is cubic when the steps on either side agree and hexagonal when they differ, so the hexagonality is $h = (1 - \langle \sigma_i \sigma_{i+1} \rangle)/2$. In the axial next-nearest-neighbour Ising (ANNNI) model of Heine and coworkers a property X of a stacking, per atom pair and relative to the cubic stacking 3C, is written with couplings J_1, J_2, J_3 between layers one, two and three apart. With the convention of Keller et al. (2023), checked against their energies: $X(2H) = 2(J_1 + J_3)$, $X(4H) = J_1 + 2J_2 + J_3$, $X(6H) = (2/3)(J_1 + 2J_2 + 3J_3)$.

Proposition. (i) A property X of the polytypes 3C, 6H, 4H and 2H is a straight line in h, $X = s \cdot h$, exactly when $J_2 = J_3 = 0$ (and then $J_1 = s/2$). (ii) For such a property the stacking faults of a cubic crystal stand X(ISF) : X(ESF) : X(TSF) = 2 : 2 : 1, where, in the same model, $X(ISF) = 4(J_1 + J_2 + J_3)$, $X(ESF) = 4(J_1 + 2J_2 + 2J_3)$ and $X(TSF) = 2(J_1 + 2J_2 + 3J_3)$.

Proof. (i) The three equations have the matrix with rows (2, 0, 2), (1, 2, 1), (2/3, 4/3, 2), of determinant $16/3 \neq 0$, so the couplings are fixed by the three values. $X = s \cdot h$ gives $X(2H) = s$, $X(4H) = s/2$, $X(6H) = s/3$, which $J_1 = s/2$, $J_2 = J_3 = 0$ satisfies, and by uniqueness only it. Conversely $J_2 = J_3 = 0$ gives $2J_1$, J_1, $(2/3)J_1$, which is $s \cdot h$ with $s = 2J_1$. (ii) With $J_2 = J_3 = 0$ the three fault values are $4J_1$, $4J_1$ and $2J_1$. ∎

So the ideal pieces of #66, which obey the law, have only a nearest coupling, and #67's fault ratios (a twin turns one layer hexagonal, an intrinsic and an extrinsic fault two each) are this 2 : 2 : 1. In a real crystal, J_2 and J_3 measure how far a property bends away from the line.

3. Volumes

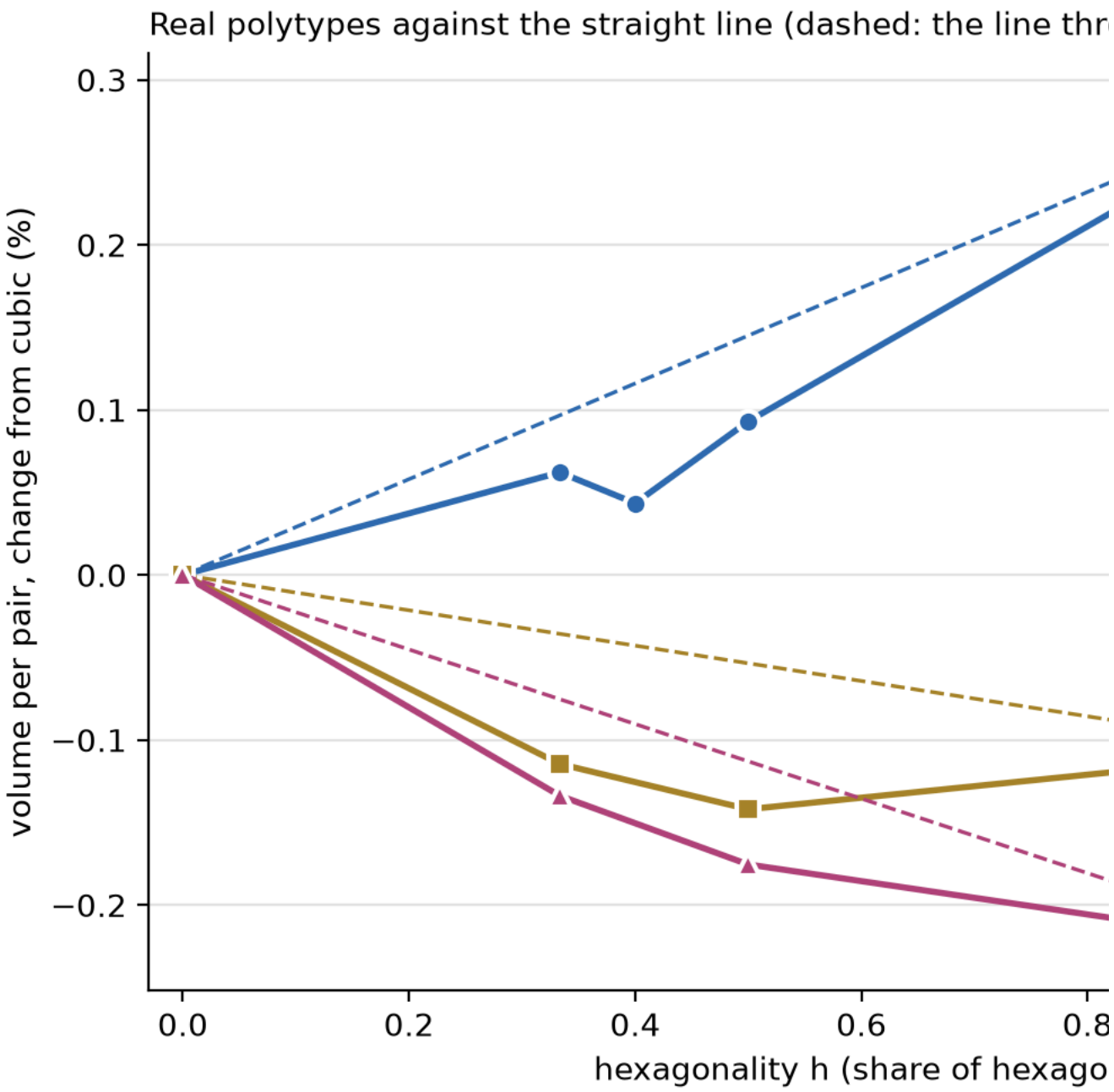


Figure 1. Volume per atom pair of real polytypes, as the change from the cubic stacking, against the hexagonality; dashed, the straight line through the two pure stackings. Silicon carbide (density-functional calculation with van der Waals correction, Kawanishi and Mizoguchi): within the precision of the published numbers (about 0.05%). Silicon and germanium (PBEsol, Keller et al.): a clear bend of up to about 0.04%.

Silicon carbide (2H, 4H, 15R, 6H, 3C). Kawanishi and Mizoguchi (2015) list measured lattice constants from six sources and three calculated sets. The volume per Si–C pair is $V = (\sqrt{3}/2)a^2(c/p)$. In every calculated set it lies on a straight line in h to within the rounding of the published numbers (largest departure 0.008–0.011 Å³ against a rounding of about 0.011 Å³, on about 20.6 Å³). The measured values differ between laboratories by up to 0.13 Å³, more than the whole change from cubic to hexagonal (under 0.3%), so they neither confirm nor refute the line.

Silicon and germanium (2H, 4H, 6H, 3C), the diamond framework of #60's pyrochlore run and of ice. Keller et al. (2023) give volumes per pair to three decimals. They do not lie on a straight line: silicon departs by up to 0.019 Å³ (on 40.1 Å³), about forty times the rounding, with its smallest volume at 4H, as the authors note; germanium by up to 0.016 Å³. Solving for the volume couplings:

Data set	J_1 (Å³/pair)	J_2	J_3	J_2/J_1, volume	J_2/J_1, energy
Si, PBEsol	−0.0155	−0.0178	−0.0060	+1.14	−0.23
Si, their ref. [46]	−0.0297	−0.0132	+0.0012	+0.45	−0.25
Ge, PBEsol	−0.0457	−0.0143	−0.0058	+0.31	−0.06

The volume couplings beyond the nearest are of the size of the nearest one, relatively larger than the energy couplings and of the opposite sign; the two silicon sets disagree on their size, so the robust statement is their order of magnitude: real relaxation adds layer couplings of

about 0.01–0.02 Å³ per pair, which the ideal geometry of the law does not have.

4. Stacking-fault energies

Keller et al. give the fault energies (mJ/m²) from their energy couplings. Against the ideal 2 : 2 : 1 of the Proposition:

Data set	intrinsic : extrinsic : twin
Ge, PBEsol (69.4, 63.3, 30.8)	2.25 : 2.06 : 1
Ge, their ref. [46]	2.32 : 2.07 : 1
Si, PBEsol (36.5, 21.2, 8.9)	4.10 : 2.38 : 1
Si, their ref. [46]	4.68 : 2.45 : 1

Germanium, whose energy coupling J_2 is small, is close to the ideal; silicon, with a larger J_2, is far from it. Measured fault energies of silicon (intrinsic about 55–69 mJ/m², extrinsic about 60, cited there) are higher than the calculated ones; the authors note that they are hard to measure.

5. Band gaps

The band gap of the silicon carbide polytypes rises with the hexagonality (Choyke, Hamilton and Patrick, 1964): about 2.36 eV (3C), 3.02 (6H), 3.23 (4H) and 3.33 (2H). The straight line through these misses them by up to 0.2 eV, and the gap couplings are $J_1 = 0.43$, $J_2 = 0.19$, $J_3 = 0.06$ eV ($J_2/J_1 = 0.45$): the classical "linear" law is a rough trend, as later work found (Matsushita and Oshiyama, 2014). The band gap is an electronic property, not a share of space, and we compare it here only for its form.

6. A metal: cerium

At room temperature cerium exists both as cubic γ-cerium (fcc, a = 5.1612 Å, h = 0) and as double-hexagonal β-cerium (dhcp, ABAC, a = 3.6811 Å, c = 11.857 Å, h = 1/2). Their volumes per atom are 34.37 and 34.79 Å³: the double-hexagonal form is 1.2% larger. With two stackings a line cannot be tested, but the size of the change from cubic to half hexagonal is thirty times that of silicon carbide or silicon.

7. Conclusion

In the language of stacking physics the Hexagonality Law is the statement that a property has only a nearest coupling between layers, and its fault ratios are then exactly 2 : 2 : 1. Real crystals follow the straight line to within their data in silicon carbide and to about 0.04% in silicon and germanium, whose small bend is a relaxation coupling of the size of the nearest one; germanium's fault energies are close to 2 : 2 : 1, silicon's are not; band gaps bend much more. The law's straight line is the ideal part of every stacking, and the departures of real crystals from it are what they add.

8. Prior art and credit

The ANNNI description of polytypes is Heine's (Cheng, Needs and Heine, 1988); the fault-energy formulas and all numbers used are those of the sources below. That a property is linear in the hexagonality exactly when its couplings beyond the nearest vanish is a direct consequence of the model, which we state with its proof for completeness; we did not find it stated in this form, nor its link to the volumes of the weighted pieces of #66. The Hexagonality Law is #66; random stackings and faults are #67.

Sources checked (28 October 2026). S. Kawanishi and T. Mizoguchi, Effect of van der Waals interactions on the stability of SiC polytypes, arXiv 1512.04626 (Table I, read; its experimental values from H. Schulz and K. H. Thiemann 1979, R. W. G. Wyckoff 1963, M. Stockmeier et al. 2009, Y. M. Tairov and V. F. Tsvetkov, A. H. Gomes de Mesquita 1967, Landolt–Börnstein 1982). M. Keller, A. Belabbes, J. Furthmüller, F. Bechstedt and S. Botti, First-principles insight in structure-property relationships of hexagonal Si and Ge polytypes, arXiv 2302.10822 (Tables I, II and V, read). Wikipedia, Polymorphs of silicon carbide (band gaps to one decimal; two-decimal values from search summaries of the literature); Wikipedia, Cerium (lattice constants at 20 °C). Search summaries, not read in full: Matsushita and Oshiyama (arXiv 1404.3588); Phys. Rev. B 89, 245308 (2014; ZnS, ZnSe, ZnTe polytypes); the RRUFF wurtzite entry; reviews of the γ–β transition of cerium. Zinc sulfide was dropped: no open, consistent table of its polytypes was found. Reference library: ask-library "lattice parameters of silicon carbide polytypes 3C 4H 6H 2H measured" (nothing relevant).

Data and code. Study 2026-10-28-measured-polytypes (sic_linear.py, sige_linear.py, annni_volume.py, gaps_faults_ce.py, fig68.py and their outputs), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Random Stackings and Stacking Faults (#67), 10.5281/zenodo.23718636.